

# Day 6 – Detecting Expired Keys

## (Management Command)

### Setup & Prerequisites

1. Setup Checklist:
- **Virtual Environment active:** Ensure your terminal shows `(.venv)` in the prompt.
  - **Django Server running:** Run `python manage.py runserver` to ensure settings are active.
  - **IDE Ready:** Open your project directory in the IDE, ready to create new nested directories and python scripts.
2. Prerequisites:
- You must have completed Day 5 successfully (Order database tables migrated, and order creation view operational).

### Learning Objectives

- By the end of this class, you will be able to:
- Explain Django custom management commands and their use cases.
  - Scaffold the specific package structure required for custom Django commands.
  - Subclass `BaseCommand` to write command line utilities.
  - Perform high-performance database batch updates using the ORM `.update()` method.
  - Construct timezone-aware database queries using the `__lte` (less than or equal to) lookup.
  - Configure cron jobs to run custom commands on a schedule.

### Classroom Timeline (45 min)

| Segment                      | Duration | Focus                                                                                                     |
|------------------------------|----------|-----------------------------------------------------------------------------------------------------------|
| 1. Hook & Big Picture        | 8 min    | Define background tasks. Explain when to use management commands vs views or tasks.                       |
| 2. Key Concepts & Core Ideas | 8 min    | Explain <code>BaseCommand</code> methods, stdout redirection, batch ORM operations, and timezone filters. |

|                                          |        |                                                                                                                               |
|------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>3. Live Coding Walkthrough</b>        | 21 min | Create directories and files, implement <code>check_expired_keys</code> logic, run manually, and discuss cron configurations. |
| <b>4. Check for Understanding</b>        | 5 min  | Q&A on <code>stdout</code> vs <code>print</code> , <code>.update()</code> side effects, and <code>crontab</code> syntax.      |
| <b>5. Class Wrap-up &amp; Checkpoint</b> | 3 min  | Verify expired keys are successfully processed by running the command in the terminal.                                        |



## Lecture Step-by-Step Delivery Plan

### 1. Hook & Big Picture (8 min)

So far, we have built APIs that run when a user makes a request (like buying a key). But who updates a key when it expires? Does the user have to request an endpoint to expire it? No, the system must check in the background automatically.

- **Q&A:**

- **Question:** Who updates a key when it expires? Does the user have to request an endpoint to expire it?
  - **Answer:** No, the system must check expirations automatically in the background using an automated script (management command) scheduled via a task runner or cron.
- **Question:** When are management commands the right tool vs. a scheduled job vs. a cron script vs. a Celery task?
  - **Answer:** Management commands are best for manual administrative scripts or commands run from the CLI. Cron jobs are standard OS-level task schedulers (perfect for triggering management commands periodically). Celery tasks are best for immediate background processing off the main thread or persistent queues.

- **Management Commands:** Django commands are scripts that can be invoked via `python manage.py [command]`. They are perfect for background scripts triggered by the OS scheduler (like cron).

### 2. Key Concepts & Core Ideas (8 min)

Introduce these command line concepts:

- **BaseCommand:** The base class for all Django CLI scripts. Custom commands must subclass it and implement a `handle()` method.
- **Stdout and Stderr Redirection:** Why commands must print output using `self.stdout.write()` instead of standard Python `print()`. Using `self.stdout.write()` integrates with logging and output redirection, enabling unit tests to capture output and allowing outputs to be piped into log files in production.

- `self.style.SUCCESS()`: Colorizes terminal logs (e.g. green for success, red for errors) for better visual feedback.
  - **Bulk Database Updates:** Contrast looping over matching rows and calling `.save()` (creates `N` SQL queries) with `.update()` (creates `1` SQL query).
  - **Timezone-aware datetime lookups:** Why using `timezone.now()` is mandatory when querying timezone-aware fields, and how `expires_at__lte` translates to SQL `expires_at <= NOW()`.
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### 3. Live Coding Walkthrough (21 min)

#### Step 3.1: Scaffolding the Directory Structure

- **Concept:** Django scans directories to discover custom commands. We must create a specific nested folder layout under our custom app. Note that every directory must contain an empty `__init__.py` file so Python treats them as packages.
- **Code:** Create the folder tree in the terminal:

```
games/  
  management/  
    __init__.py  
  commands/  
    __init__.py  
    check_expired_keys.py
```

- **Verify:** Confirm that both `__init__.py` files are created. ⚠ **Common Pitfalls:**
- **Missing `__init__.py`:** If either package initialization script is missing, Django will fail to register the command, returning `Unknown command: 'check_expired_keys'`.

#### Step 3.2: Implementing the Command Logic

- **Concept:** Create `games/management/commands/check_expired_keys.py`. Write a command class that queries all active keys whose expiration datetime is in the past, changes their status to `expired`, and prints a count of affected keys.
- **Code:** Write the following content inside `games/management/commands/check_expired_keys.py`:

```
from django.core.management.base import BaseCommand  
from django.utils import timezone  
from games.models import GameKey  
  
# Custom command class MUST inherit from BaseCommand  
class Command(BaseCommand):  
    # Help text displayed when running: python manage.py check_expired_keys --help  
    help = 'Mark active keys whose expiry has passed as expired.'  
  
    # The handle method is the core execution logic invoked when running the command
```

```
def handle(self, *args, **options):
    # expires_at__lte (Less Than or Equal to) filters for expirations in the past
    expired_keys = GameKey.objects.filter(
        expires_at__lte=timezone.now(),
        status='active'
    )

    # .update() modifies all queried rows with a single SQL query:
    # UPDATE games_gamekey SET status='expired' WHERE expires_at <= NOW() AND status='active'
    count = expired_keys.update(status='expired')

    # self.stdout.write displays outputs in the console. self.style.SUCCESS makes it green
    self.stdout.write(self.style.SUCCESS(f'Expired {count} keys.'))
```

- **Verify:** Ensure the file compiles without syntax errors.

### Step 3.3: Manual CLI Testing

- **Concept:** We will manually trigger a key expiration. Log into the Django Admin dashboard and create or modify a `GameKey` record so that its `expires_at` value is set to one hour in the past, and its status is set to `active`. Then, run the command.
- **Code:** Execute the management command:

```
python manage.py check_expired_keys
```

- **Verify:** Check the console output. It should output a success message (e.g. `Expired 1 keys.`). Reload the Admin page and verify the status of the expired key has changed to `Expired`. ⚠ **Common Pitfalls:**
- **Using `datetime.now()` instead of `timezone.now()`:** If you use naive Python datetimes, Django will raise a `TypeError` due to comparing offset-naive and offset-aware datetimes.
- **Queryset caching after updates:** Note that `.update()` modifies the rows in the database, but does not update active Python model objects in memory. If we reference the objects inside a cached queryset after calling `.update()`, they will still show the old status.

### Step 3.4: Scheduling in Production via Cron

- **Concept:** To run this script every 15 minutes automatically on our production Linux server, we would add a line to our system crontab file pointing to the `virtualenv` python interpreter.
- **Code:** Open crontab config (via `crontab -e`) and add:

```
*/15 * * * * /path/to/.venv/bin/python /path/to/manage.py check_expired_keys
```

- **Verify:** Discuss the cron parameters: `*/15` means every 15 minutes, followed by hour, day of month, month, and day of week wildcards.

## 4. Check for Understanding (5 min)

**Question:** "Why do we use `self.stdout.write()` instead of `print()` inside a management command?"

**Answer:** Using `self.stdout.write()` is best practice because it integrates with Django's internal logging systems and output redirection. In testing, it allows the output to be intercepted and asserted on (using `call_command`), whereas `print()` writes directly to the standard system `stdout`, making it harder to test or suppress.

**Question:** "What's the performance difference between `.update()` and looping + `.save()`? Are there situations where you'd prefer `.save()` anyway?"

**Answer:** `.update()` executes a single SQL `UPDATE` statement in the database, which is extremely fast and efficient for bulk operations. Looping and calling `.save()` executes a separate SQL query for each record ( $N$  queries), which is slow. However, you would prefer `.save()` if you need to trigger model `save()` overrides, pre/post-save signals, or validation, which `.update()` bypasses.

**Question:** "What does `expires_at__lte=timestzone.now()` translate to in SQL?"

**Answer:** It translates to a `WHERE` clause: `WHERE expires_at <= 'CURRENT_TIMESTAMP_VALUE'`. This filters for records where the expiration datetime is less than or equal to the current time.

**Question:** "If we want this command to run every 15 minutes automatically, what are our options?"

**Answer:** 1) Setup a system-level cron job that executes the command through the `virtualenv python`. 2) Use a Celery Beat task that runs the management command (or its logic) on a schedule. 3) Use cloud-specific schedulers (like Render Cron Jobs or AWS EventBridge) to trigger the command.

## 5. Daily Checkpoint & Wrap-up (3 min)

- **Verification Criteria:**

- ☐ The `check_expired_keys.py` script is placed under `games/management/commands/`.
- ☐ Creating a key with a past expiration date and running `python manage.py check_expired_keys` runs successfully.
- ☐ The script prints a green confirmation message to the terminal.
- ☐ The database records are updated from `active` to `expired` successfully.